

STAT 405 Project Proposal

Team Member:

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Github website:

<https://github.com/HaoyuYou/Bayesian-Statistics-Project.git>

Dataset

main: Bike Sharing Demand

The dataset used in this project is the Bike Sharing Dataset from the UCI Machine Learning Repository. It contains information about daily bike rental usage in a bike sharing system from 2011 to 2012.

We use the daily dataset (`day.csv`), which includes 731 observations, where each observation represents one day. The dataset records the total number of bike rentals (`cnt`) along with several explanatory variables related to time and weather conditions.

Key variables include temperature (`temp`), humidity (`hum`), wind speed (`windspeed`), season (`season`), weather situation (`weathersit`), and whether the day is a working day (`workingday`). These variables help explain patterns in bike rental demand.

```
bike <- read.csv("data/day.csv")
cat_vars <- c("season", "yr", "mnth", "holiday", "weekday", "workingday", "weathersit")
bike[cat_vars] <- lapply(bike[cat_vars], as.factor)
head(bike)
```

```
##      instant      dteday season yr mnth holiday weekday workingday weathersit
## 1         1 2011-01-01      1  0   1         0         6         0         2
## 2         2 2011-01-02      1  0   1         0         0         0         2
## 3         3 2011-01-03      1  0   1         0         1         1         1
## 4         4 2011-01-04      1  0   1         0         2         1         1
## 5         5 2011-01-05      1  0   1         0         3         1         1
## 6         6 2011-01-06      1  0   1         0         4         1         1
##      temp      atemp      hum windspeed casual registered cnt
## 1 0.344167 0.363625 0.805833 0.1604460    331         654  985
## 2 0.363478 0.353739 0.696087 0.2485390    131         670  801
## 3 0.196364 0.189405 0.437273 0.2483090    120        1229 1349
## 4 0.200000 0.212122 0.590435 0.1602960    108        1454 1562
## 5 0.226957 0.229270 0.436957 0.1869000     82        1518 1600
## 6 0.204348 0.233209 0.518261 0.0895652     88        1518 1606
```

URL:

<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

backup: BMW Global Automotive Sale

```
bmw <- read.csv("data/bmw_global_sales_2018_2025.csv")
head(bmw)
```

##	Year	Month	Region	Model	Units_Sold	Avg_Price_EUR	Revenue_EUR	BEV_Share
## 1	2018	1	Europe	3 Series	7822	47482	371404204	0.011
## 2	2018	1	Europe	5 Series	10280	61685	634121800	0.019
## 3	2018	1	Europe	X3	3105	58433	181434465	0.022
## 4	2018	1	Europe	X5	7420	67955	504226100	0.021
## 5	2018	1	Europe	X7	8474	92300	782150200	0.035
## 6	2018	1	Europe	i4	10607	64082	679717774	0.035
##	Premium_Share	GDP_Growth	Fuel_Price_Index					
## 1	19.12	3.5	1					
## 2	19.12	3.5	1					
## 3	19.12	3.5	1					
## 4	19.12	3.5	1					
## 5	19.12	3.5	1					
## 6	19.12	3.5	1					

URL:

<https://www.kaggle.com/dmahajanbe23/bmw-global-automotive-sales>

Scientific Question:

How do weather conditions and seasonal factors influence daily bike rental demand? Does a more flexible Bayesian count model provide a better fit and more realistic uncertainty quantification than a simpler baseline model?

More specifically, we aim to model how variables such as temperature, humidity, wind speed, season, and working day status affect the total number of bike rentals (`cnt`).

Bike sharing demand is influenced by both environmental conditions and human behavior patterns. Understanding these relationships can help improve demand prediction and support better planning for urban transportation systems.

Using Bayesian modelling allows us to quantify uncertainty in parameter estimates and make probabilistic interpretations of the effects of different predictors on bike rental demand.

Methodology:

We plan to build Bayesian models to explain and predict the number of daily bike rentals. In particular, we will consider two models: a simple baseline model and a more flexible regression model incorporating environmental predictors.

Before fitting the Bayesian models, we will perform exploratory data analysis (EDA) to understand the distributions of key variables and their relationships with bike rental demand. This includes visualizing the distribution of the response variable (`cnt`) and examining how it varies with weather conditions and seasonal factors.

Model 1: Naive Model

As a baseline, we first consider a simple Bayesian Poisson regression model for daily bike-sharing demand. Let Y_i be the total rental count (`cnt`) on day i . Since the response variable is a nonnegative count, a Poisson

model provides a natural starting point for modeling the distribution of daily demand. We assume

$$Y_i | \lambda_i \sim \text{Poisson}(\lambda_i) \quad \log(\lambda_i) = \beta_0 + \beta_1 \text{temp}_i + \beta_2 \text{hum}_i + \beta_3 \text{windspeed}_i$$

In this model, λ_i represents the expected number of bike rentals on day i , and its logarithm is modeled as a linear function of three continuous weather-related predictors: temperature, humidity, and windspeed. The log link is used to ensure that the Poisson mean λ_i is always positive.

Model 2: Complex Model

As a more flexible model, we then consider a Bayesian hierarchical Poisson regression model with month-specific intercepts. The purpose of this hierarchical structure is to capture systematic variation in baseline demand across months while allowing partial pooling across groups. This allows the model to borrow strength across months and produce more stable estimates.

For the other calendar-related factors, we treat them as fixed effects because of their small number of groups, even if some of them are relatively imbalanced. Because season is largely determined by month, including both month effects and season effects would introduce substantial redundancy and may lead to aliasing issues. Therefore, once month is used as the main grouping factor in the hierarchical model, season is omitted from the linear predictor. Therefore, for this hierarchical model, we assume

$$Y_i | \lambda_i \sim \text{Poisson}(\lambda_i),$$

$$\log(\lambda_i) = \alpha_{m[i]} + \beta_1 \text{temp}_i + \beta_2 \text{hum}_i + \beta_3 \text{windspeed}_i + \beta_4 \text{holiday}_i + \beta_5 \text{workingday}_i + \beta_6 \text{weathersit}_i + \beta_7 \text{yr}_i,$$

where $m[i] \in \{1, \dots, 12\}$ indicates the month associated with observation i . The month-specific intercepts are modeled hierarchically as

$$\alpha_m \sim N(\mu_\alpha, \sigma_\alpha^2), \quad m = 1, \dots, 12.$$

We assign weakly informative priors to the regression coefficients and hyperparameters, for example,

$$\begin{aligned} \beta_j &\sim N(0, 10^2), \quad j = 1, \dots, 7, \\ \mu_\alpha &\sim N(0, 10^2), \quad \sigma_\alpha \sim \text{HalfNormal}(0, 5). \end{aligned}$$

This hierarchical Bayesian framework allows us to model both the overall effects of weather and calendar-related predictors and the month-to-month heterogeneity in baseline demand.

Inference

The models will be implemented using the `simPPL` probabilistic programming framework used in this course. Posterior inference will be performed using MCMC sampling.

We will estimate:

- posterior distributions of model parameters
- uncertainty in predicted bike rental demand
- the relative importance of different predictors

Finally, we will compare the two models using posterior predictive checks to evaluate their ability to explain the observed data.

Contribution plan:

This is a team project and both team members will contribute to all stages of the project.

Ruohan Sun

- Data preprocessing and exploratory data analysis (EDA)
- Implementation of the Bayesian models
- Posterior inference and model diagnostics
- Visualization of results

Haoyu You

- Literature review and background research
- Model formulation and prior specification
- Model comparison and interpretation of results
- Writing and editing the final report